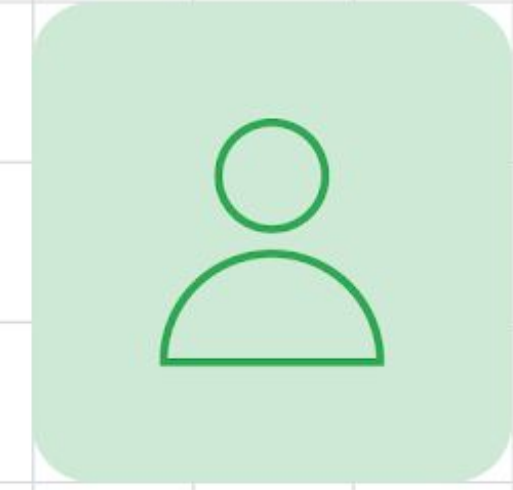
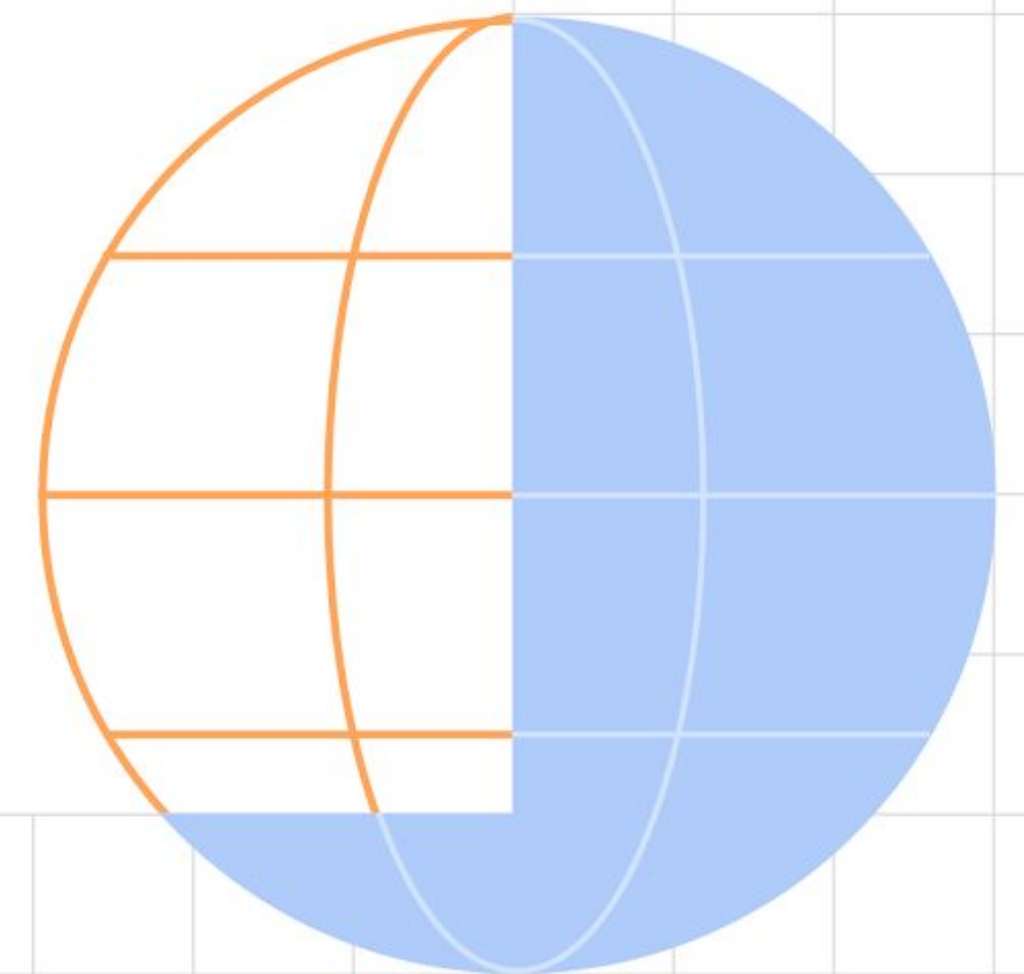




Agentic Applications Patterns - Protocols

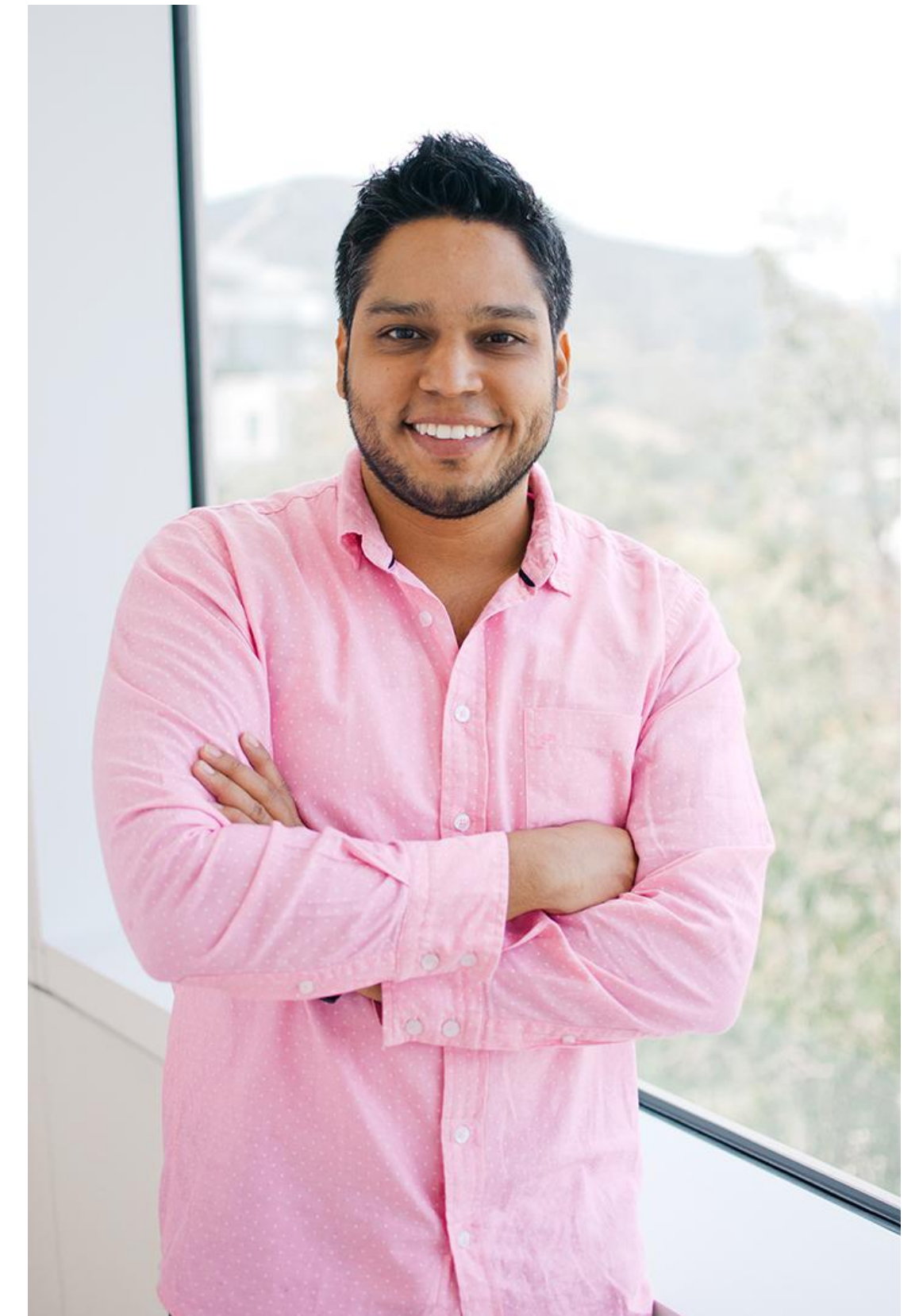


Juan Guillermo Gómez
GDE Firebase & GCP & AI/ML
@jggomezt



Juan Guillermo Gómez

- AI & Software Advisor
- Co-leader and co-founder of GDG Cali.
- Founder DevHack.
- Google Developer Expert (GDE) in Firebase & GCP & AI/ML
- BS in System Engineering
- MS in Software Engineering
- MS in AI and Data Science
- devhack.co



Networking



LinkedIn



X



Instagram



Youtube

Agenda

- ❖ What is Agent?
- ❖ Reasoning Loop
- ❖ Agentic Systems
- ❖ Patterns
- ❖ Tactics
- ❖ Protocols



Now, let's go deeper...



From LLMs to AI Agents



Are these AI Agents?

AI program designed to perform a specific task by leveraging a set of tools to accomplish it 🤔

Little AI Apps 🤔

For example, an AI program can use external tools to research a topic and generate a summary of the findings 🤔

AI Apps acting autonomous 🤔

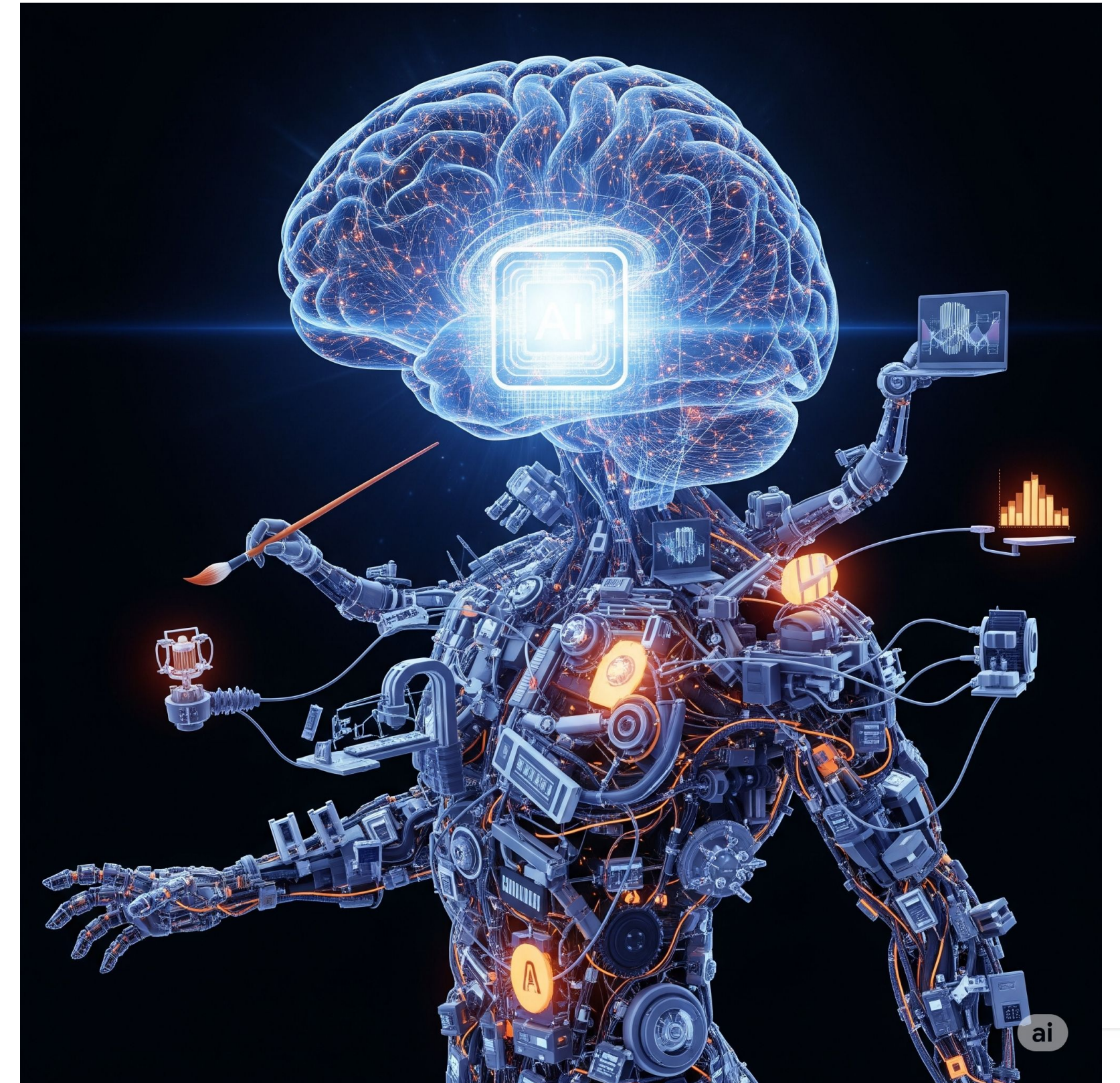
AI App must be able to "reason" 🤔

AI Agents

AI agents are software systems that use AI to pursue goals and complete tasks on behalf of users. They show reasoning, planning, and memory and have a level of autonomy to make decisions, learn, and adapt. (Google)

AI Agents

AI Agents have a goal and use a reasoning model to break it into subtasks. They control the flow, learn from errors, adapt, revise plans, and make decisions. They execute actions using tools to interact with external system. (DevHack)



AI Agents



AI Agents



AI Agents



AI Agents



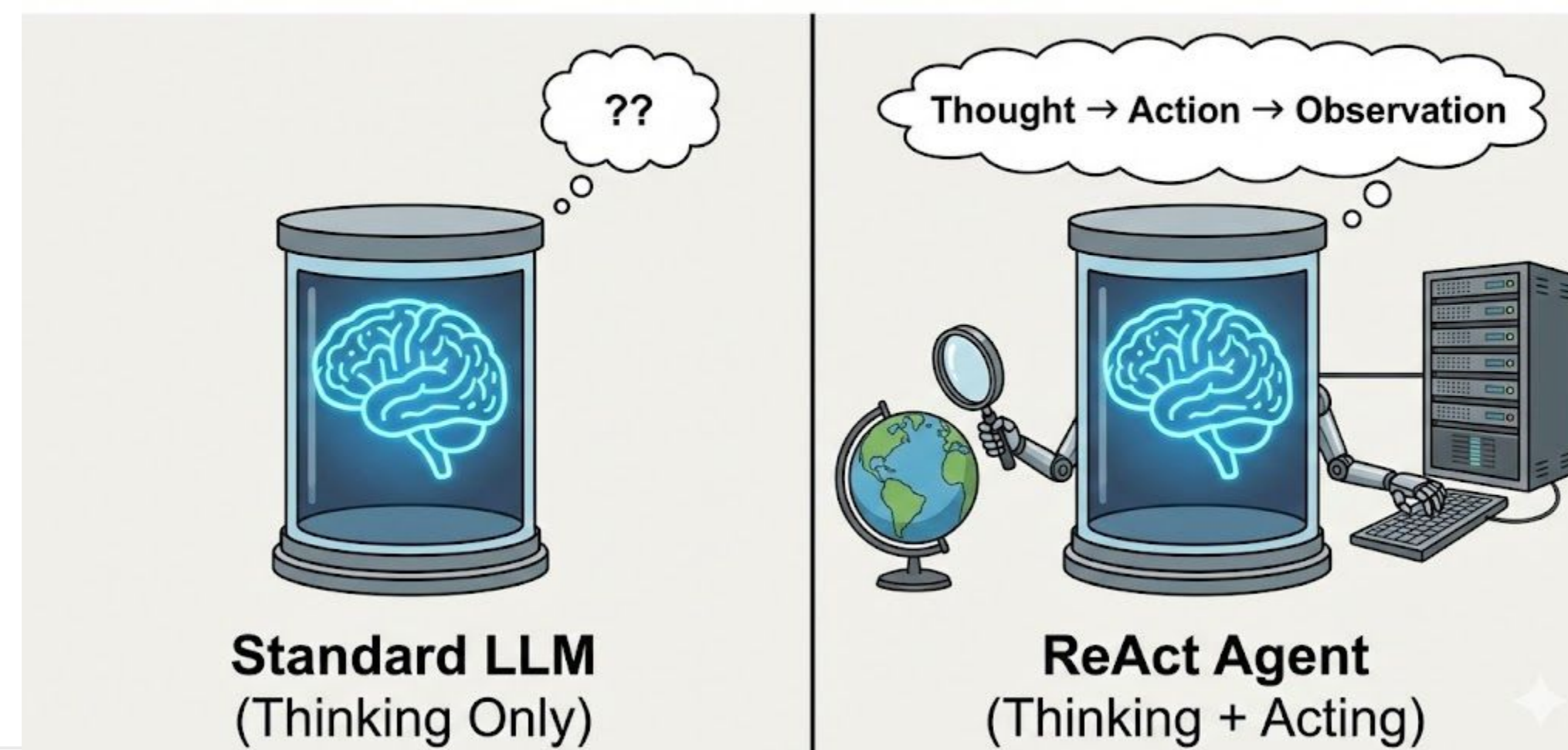
AI Agents



ReAct

- ❖ **Concept:** A paradigm that synergizes Reasoning + Acting.
- ❖ **The Shift:** Breaks the traditional Input -> Output flow. It converts the model into a dynamic loop of Think -> Act -> Observe.
- ❖ **Goal:** To enable LLMs to solve complex, multi-step problems and interact with the external world (APIs, Databases) rather than relying solely on training memory.

What is ReAct?



Agent Core Components

01	Model	The LLM powering the agent's reasoning and decision-making
02	Tools	External functions or APIs the agent can use to take action
03	Instructions	Explicit guidelines and guardrails defining how the agent behaves

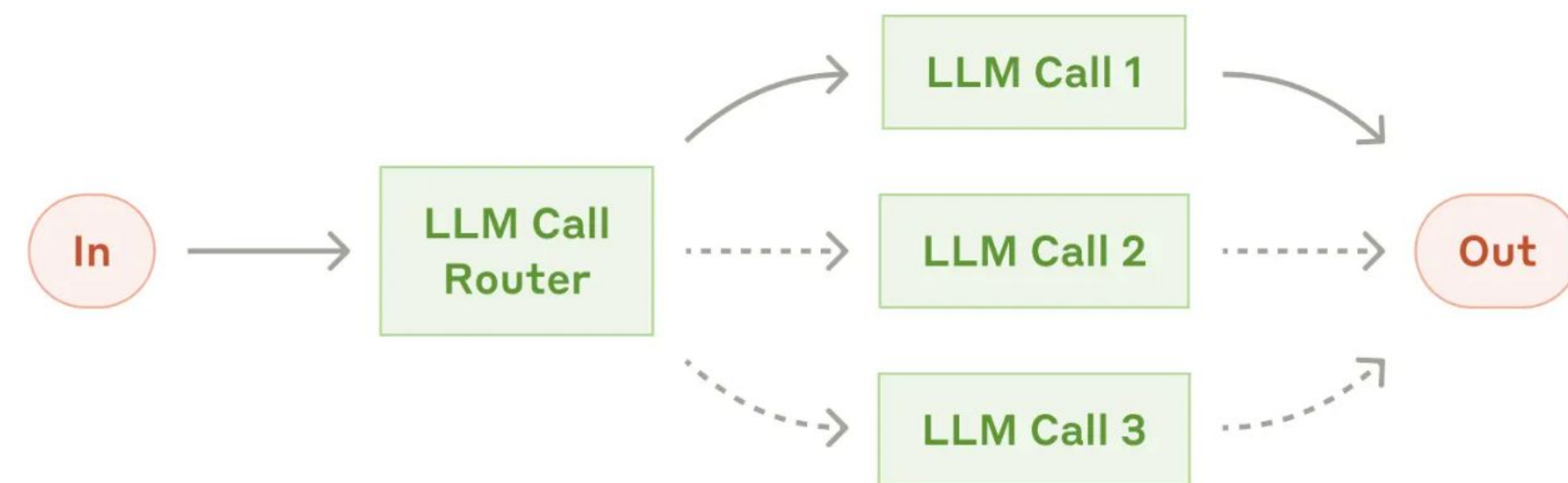
Agentic Systems



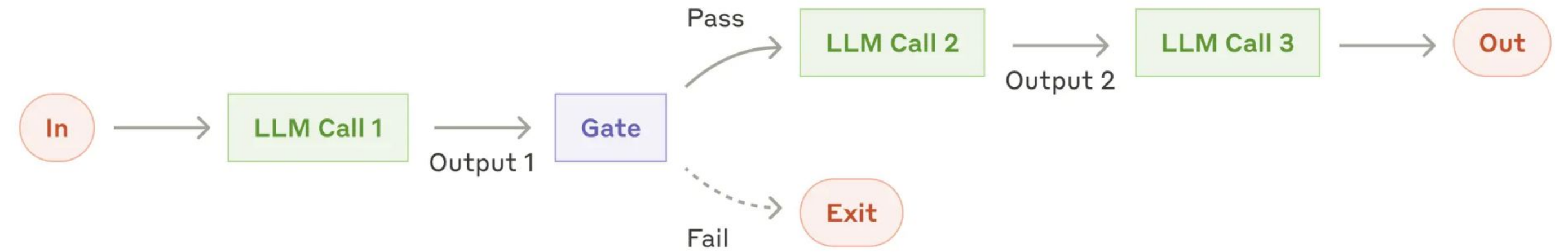
Figure 2: Agentic system in 5 steps

Workflows

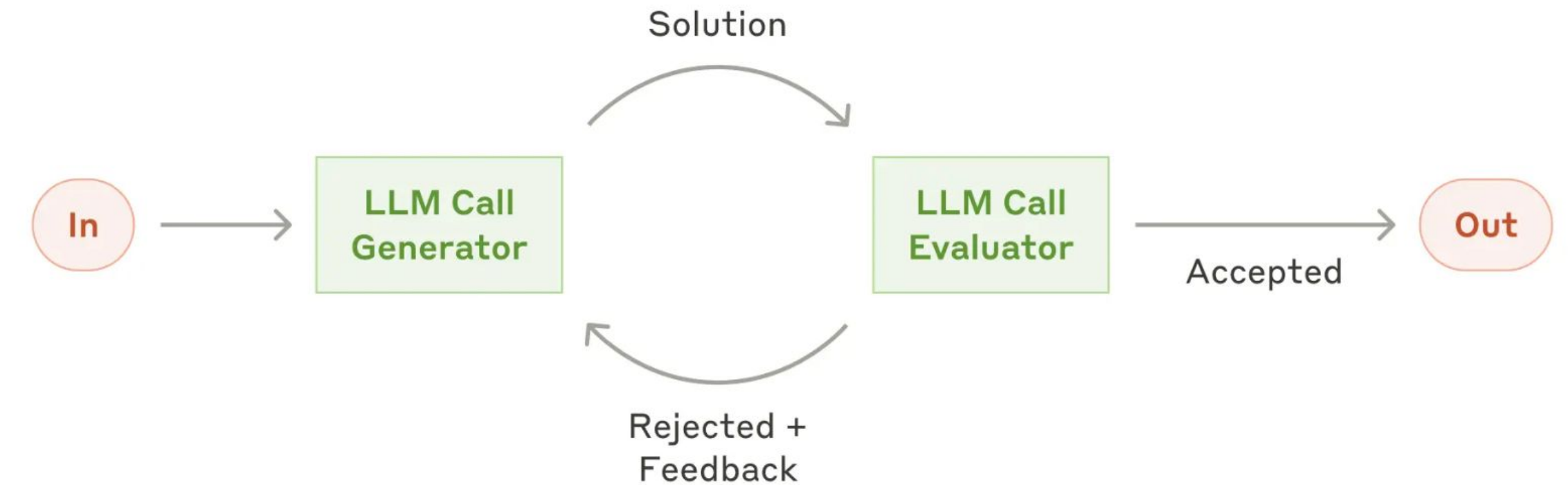
Workflow: Routing



Workflow: Prompt chaining



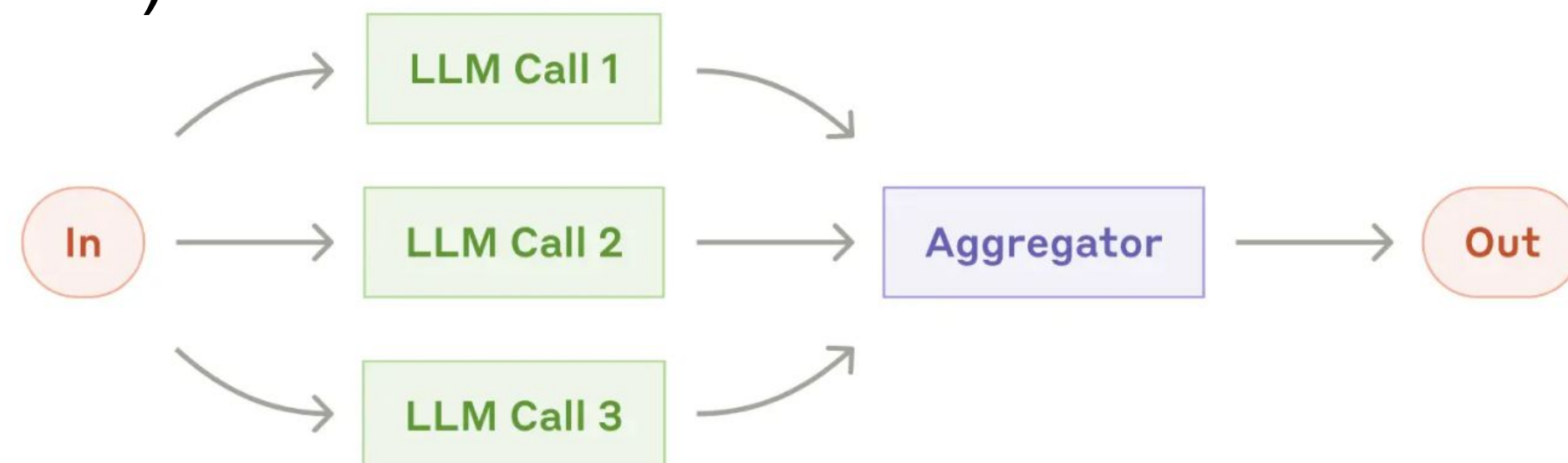
Workflow: Evaluator-optimizer (reflection)



Workflow: Orchestrator-workers (Supervisor)



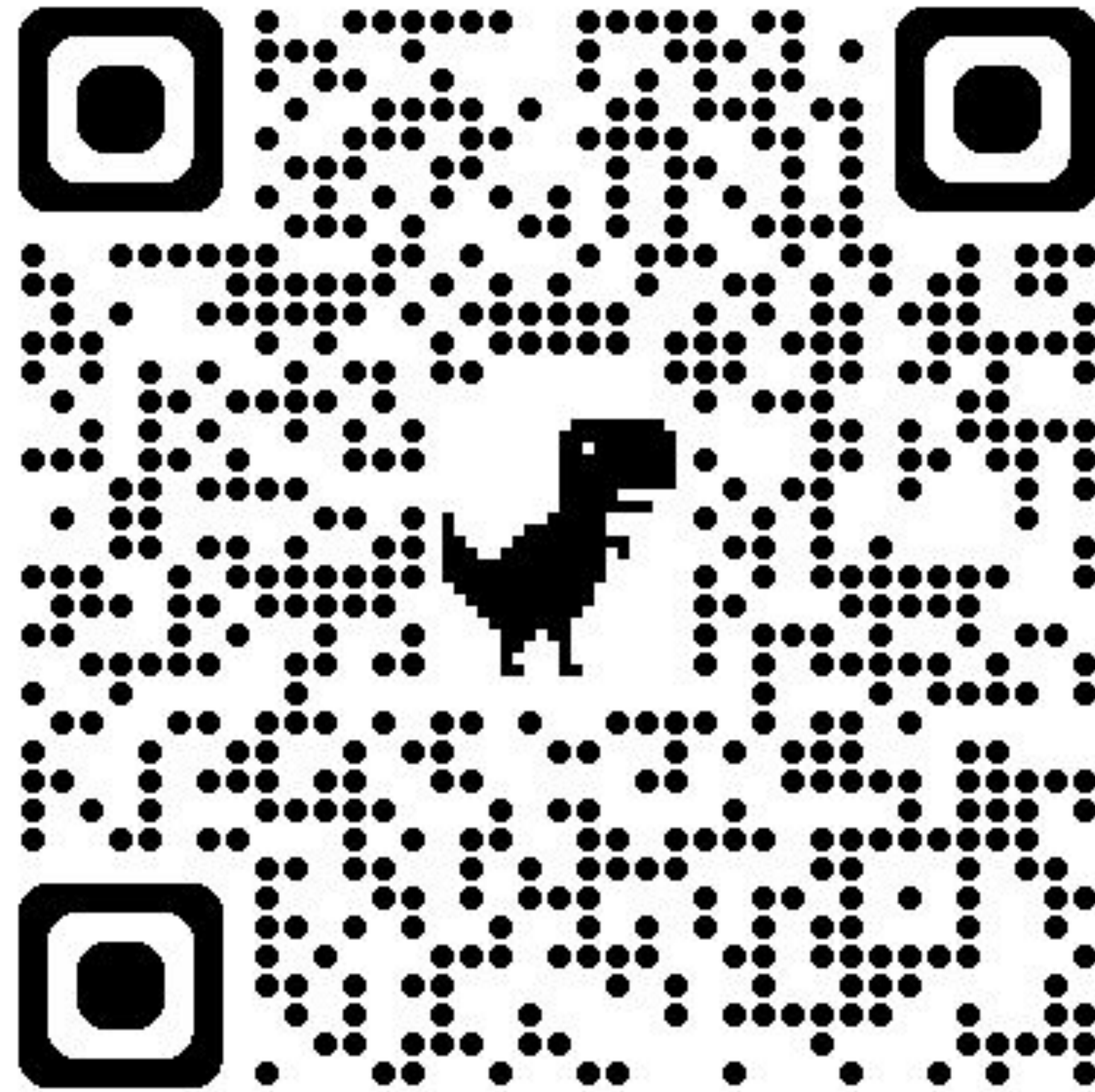
Workflow: Parallelization (Breaking a task into independent subtasks run in parallel & vote)



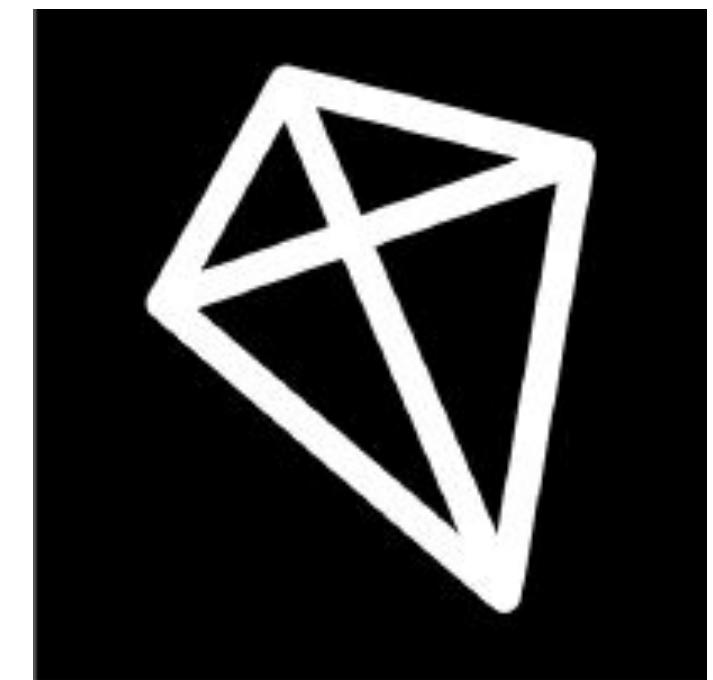
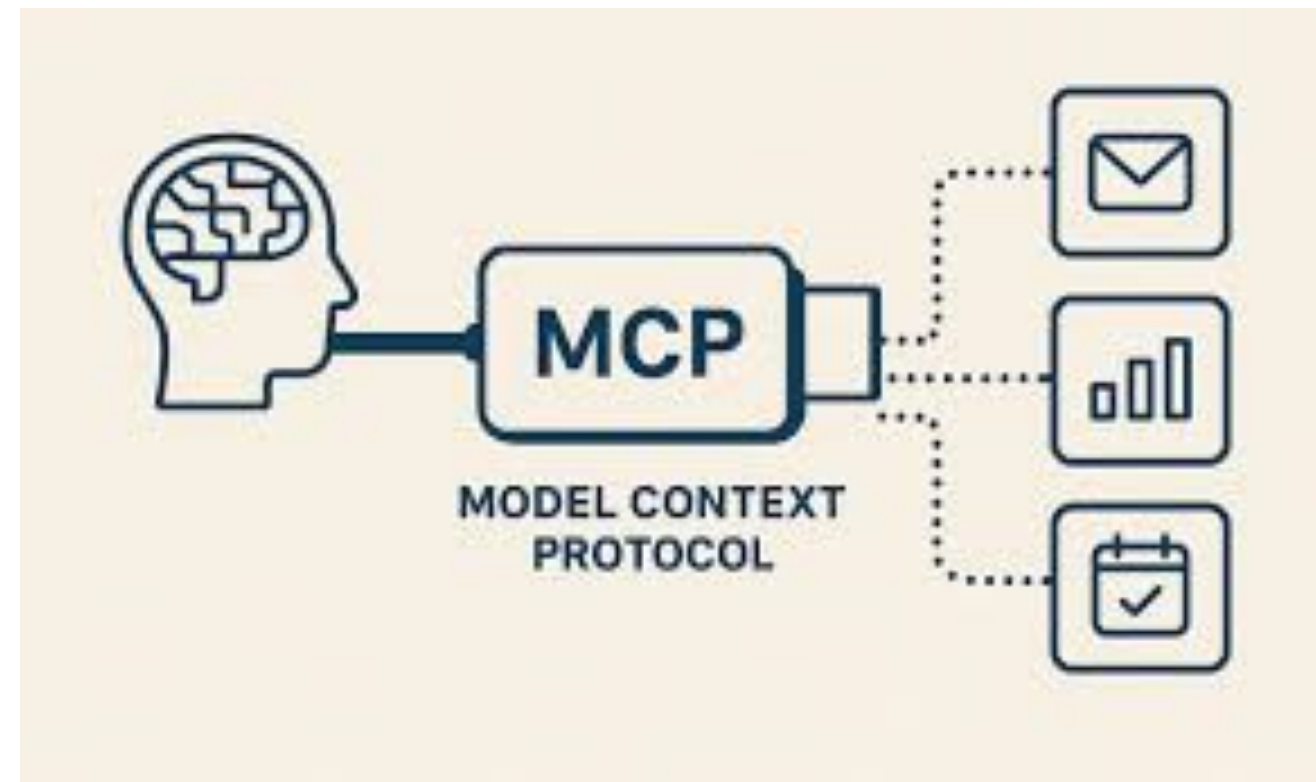
Examples



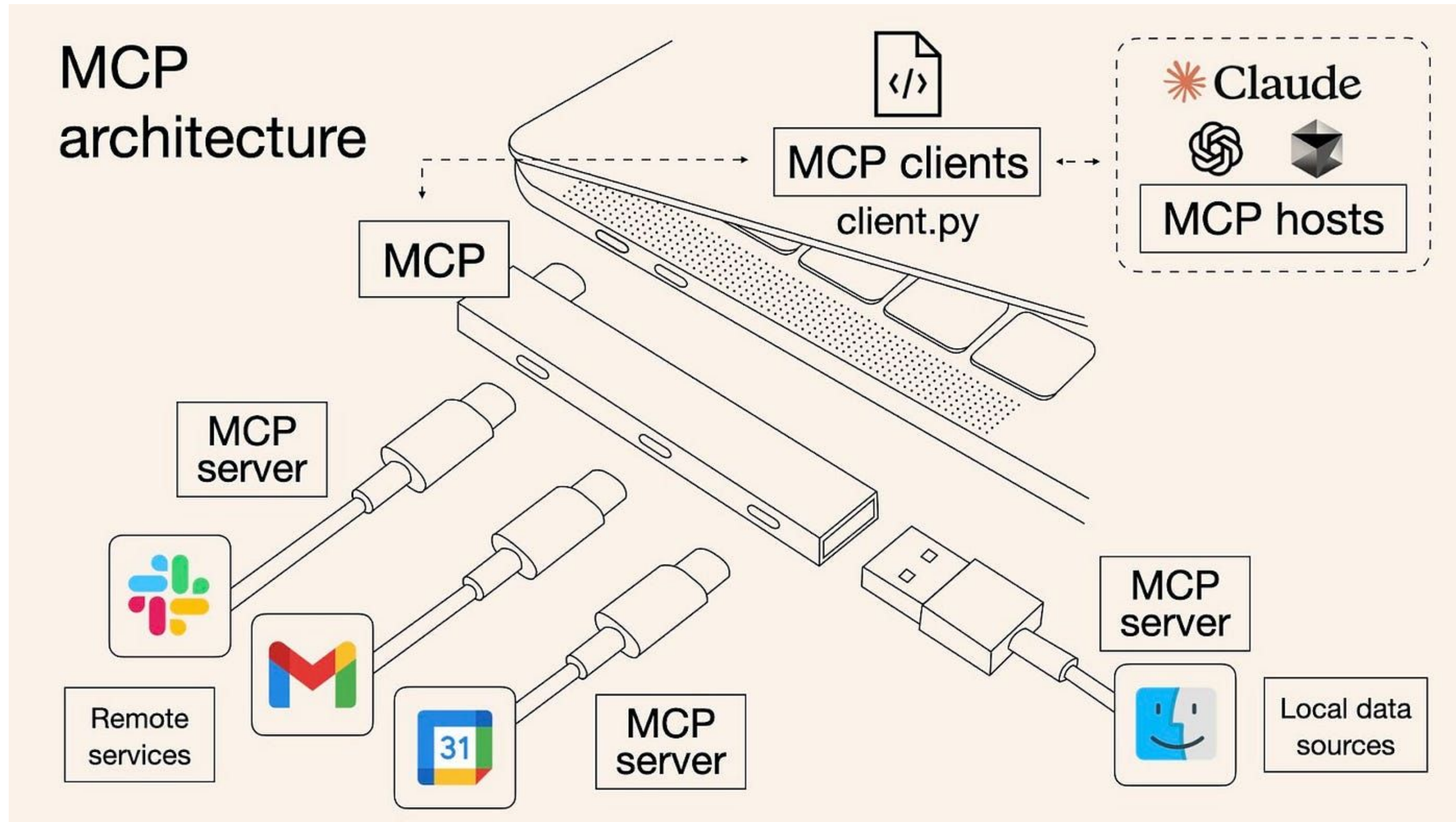
Repository



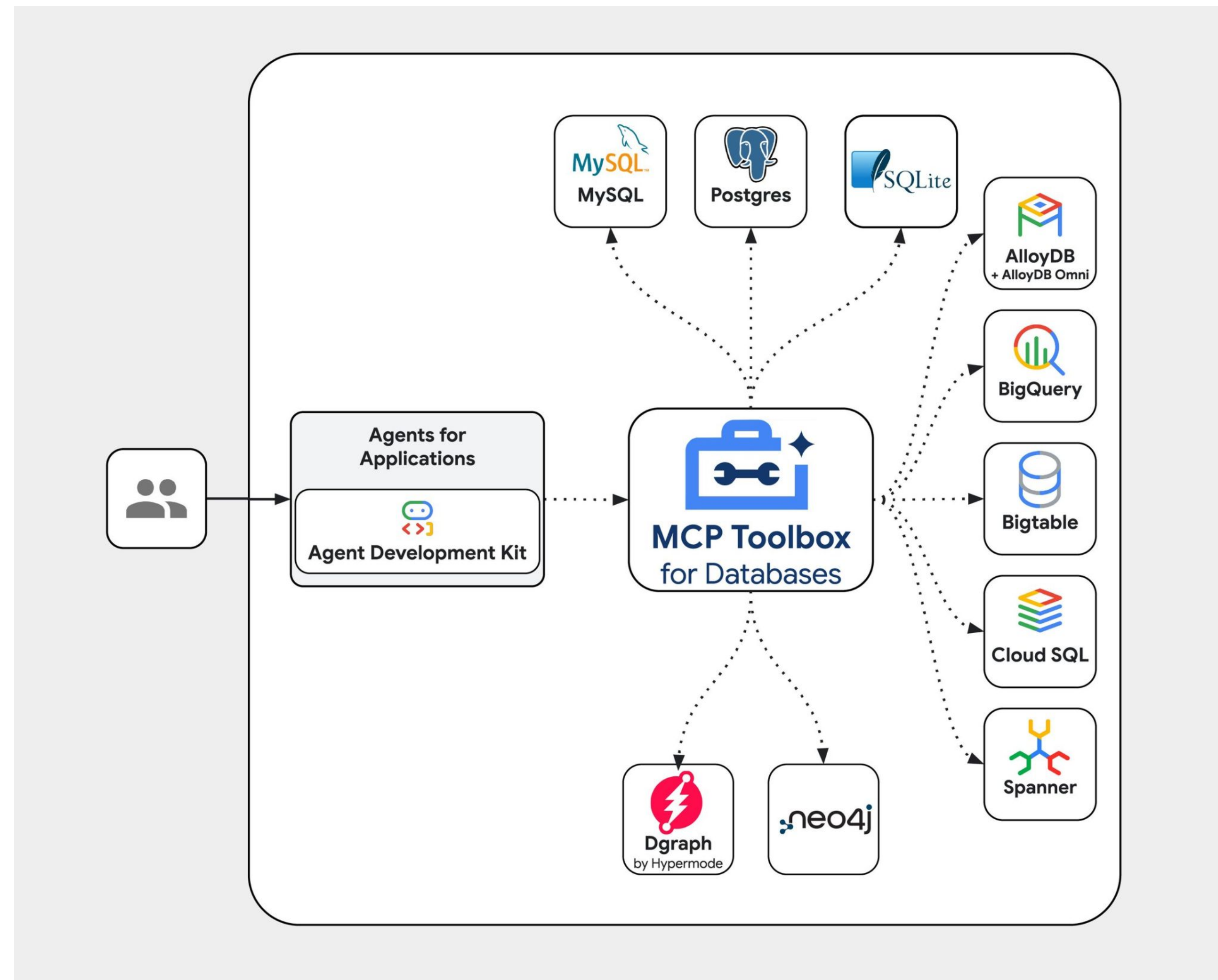
Protocols



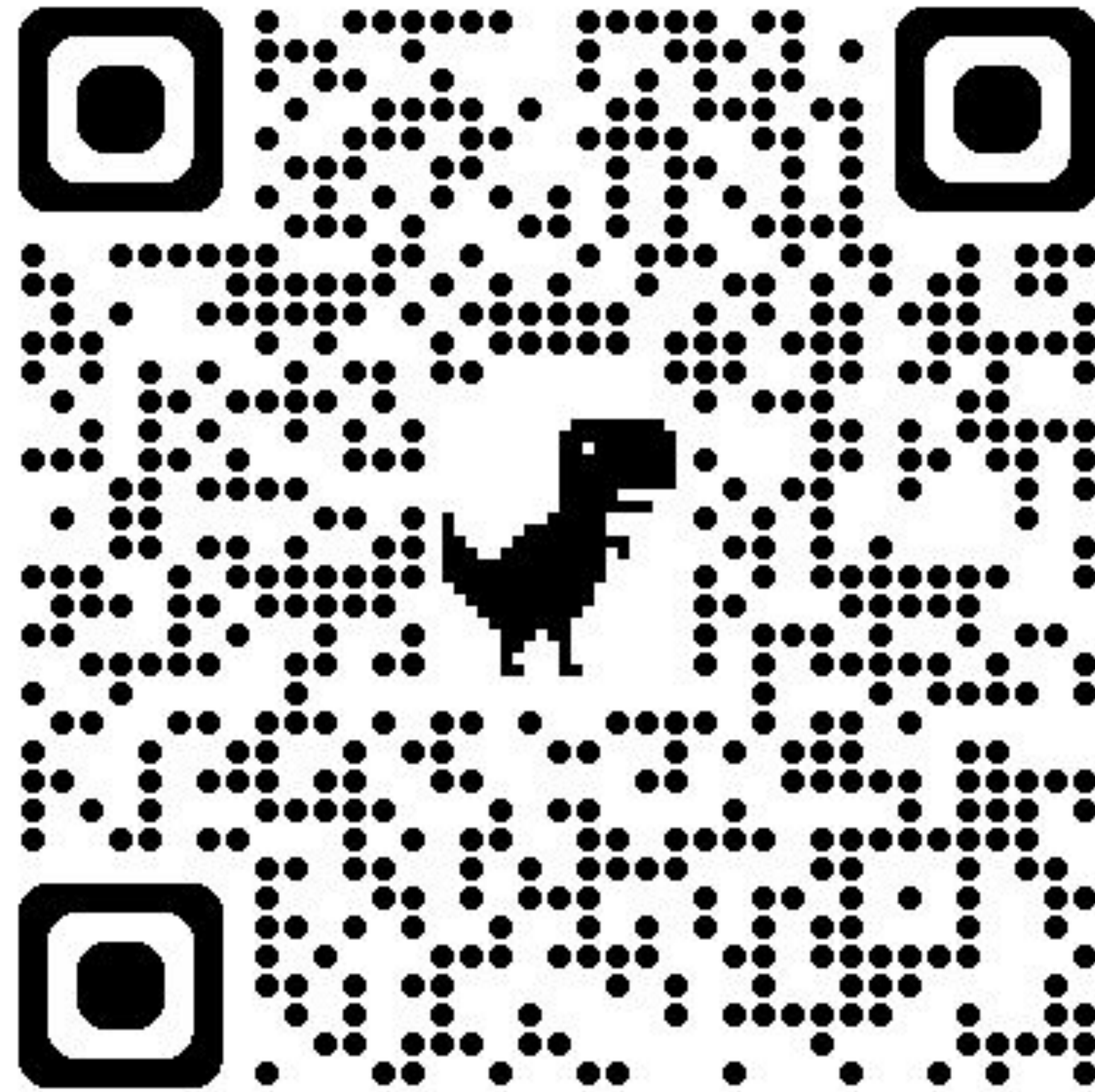
Model Context Protocol (MCP)



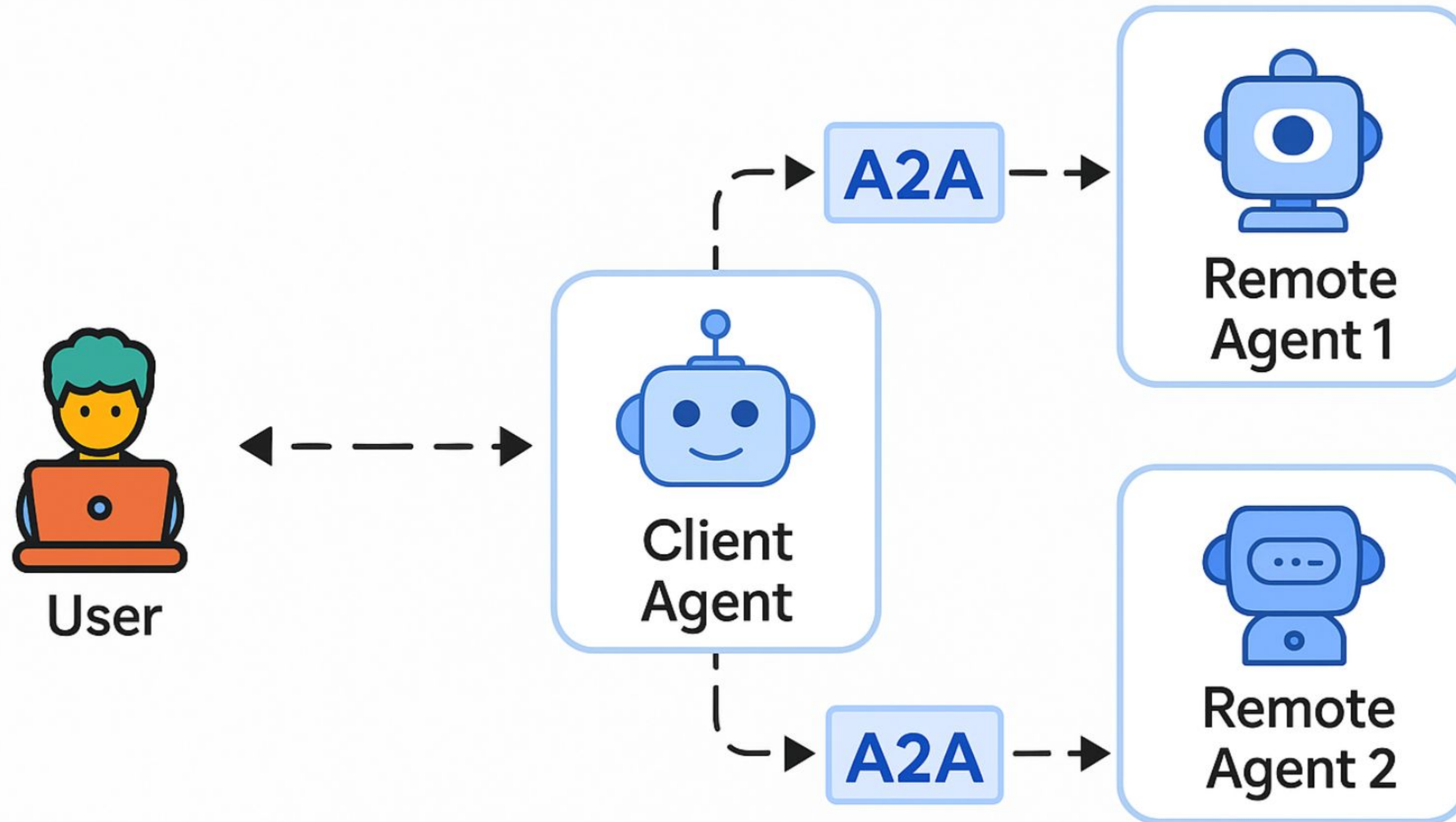
Model Context Protocol (MCP)



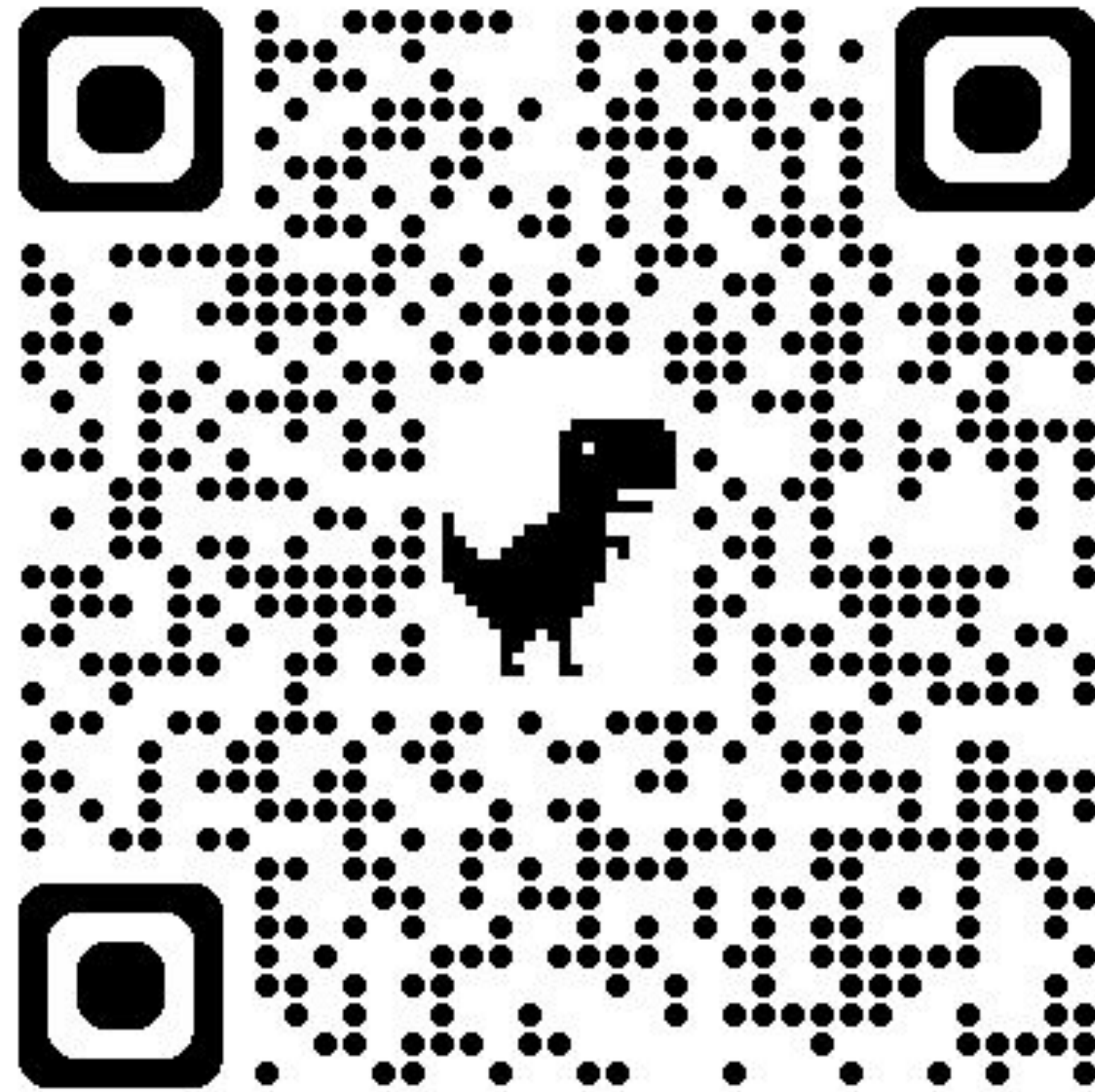
Repository



Agent 2 Agent (A2A)



Repository



A2UI

Create

Gallery

Components

Icons

Widgets

No widgets yet

Gallery

Flight Status

✈️ OS 87

Mon, Dec 15

Vienna

→

New York

Departs	Status	Arrives
10:15 AM	On Time	2:30 PM

Email Compose

FROM alex@acme.com

TO jordan@acme.com

SUBJECT Q4 Revenue Forecast

Hi Jordan,

Following up on our call. Please review the attached Q4 forecast and let me know if you have questions before the board meeting.

Best,

Alex

Send email

Discard

Calendar Day

Task Card

Review pull request

!

Review and approve the authentication module changes.

Today Backend

User Profile

Sarah Chen

@sarahchen

Product Designer at Tech Co.
Creating delightful experiences.

12.4K

892

347

Followers

Following

Posts

Follow

Login Form

Coffee Order

☕ Sunrise Coffee

Oat Milk Latte \$6.45

Grande, Extra Shot

Chocolate Croissant \$4.25

Warmed

Subtotal \$10.70

Tax \$0.96

Total \$11.66

Purchase

Add to cart

Sports Player Card

Marcus Johnson

#23 LA Lakers

28.4

7.2

6.8

P.P.G.

R.P.G.

A.P.G.

Track List

☰ Focus Flow

1 Weightless 8:09

2 Clair de Lune 5:12

3 Ambient Light 6:45

Brian Eno

Recipe Card

Mediterranean Quinoa Bowl

★ 4.9 (1,247 reviews)

15 min prep

20 min cook

Serves 4

Software Purchase Form

Purchase License

Design Suite Pro

Number of seats 10 seats

Billing period Annual

Total \$1,188/year

Confirm Purchase

Cancel

Contact Card

David Park

Engineering Manager

+1 (555) 234-5678

david.park@company.com

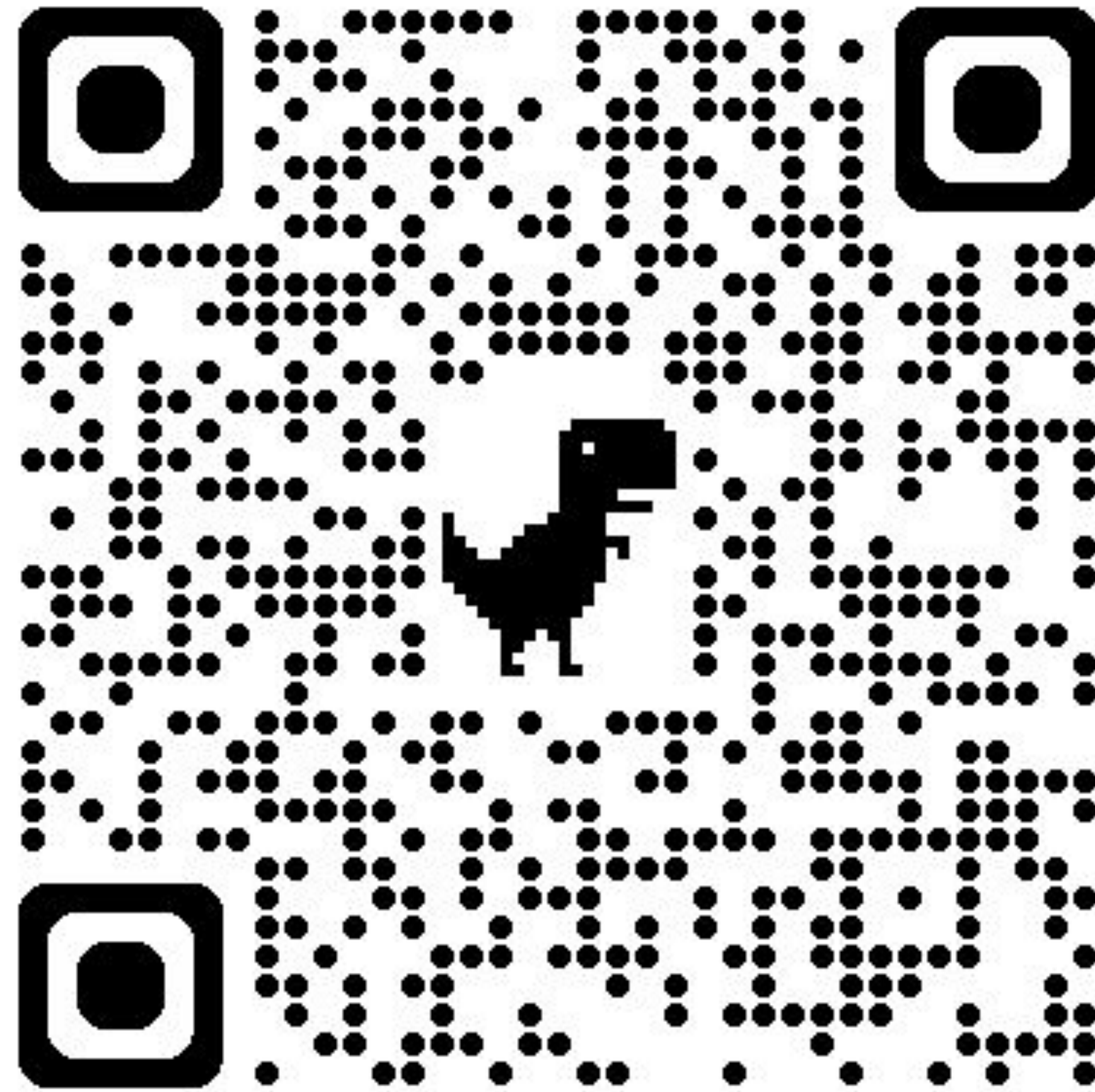
San Francisco, CA

Call

Message

Experts

Repository



A2UI

A2UI



AI agent



Declarative message

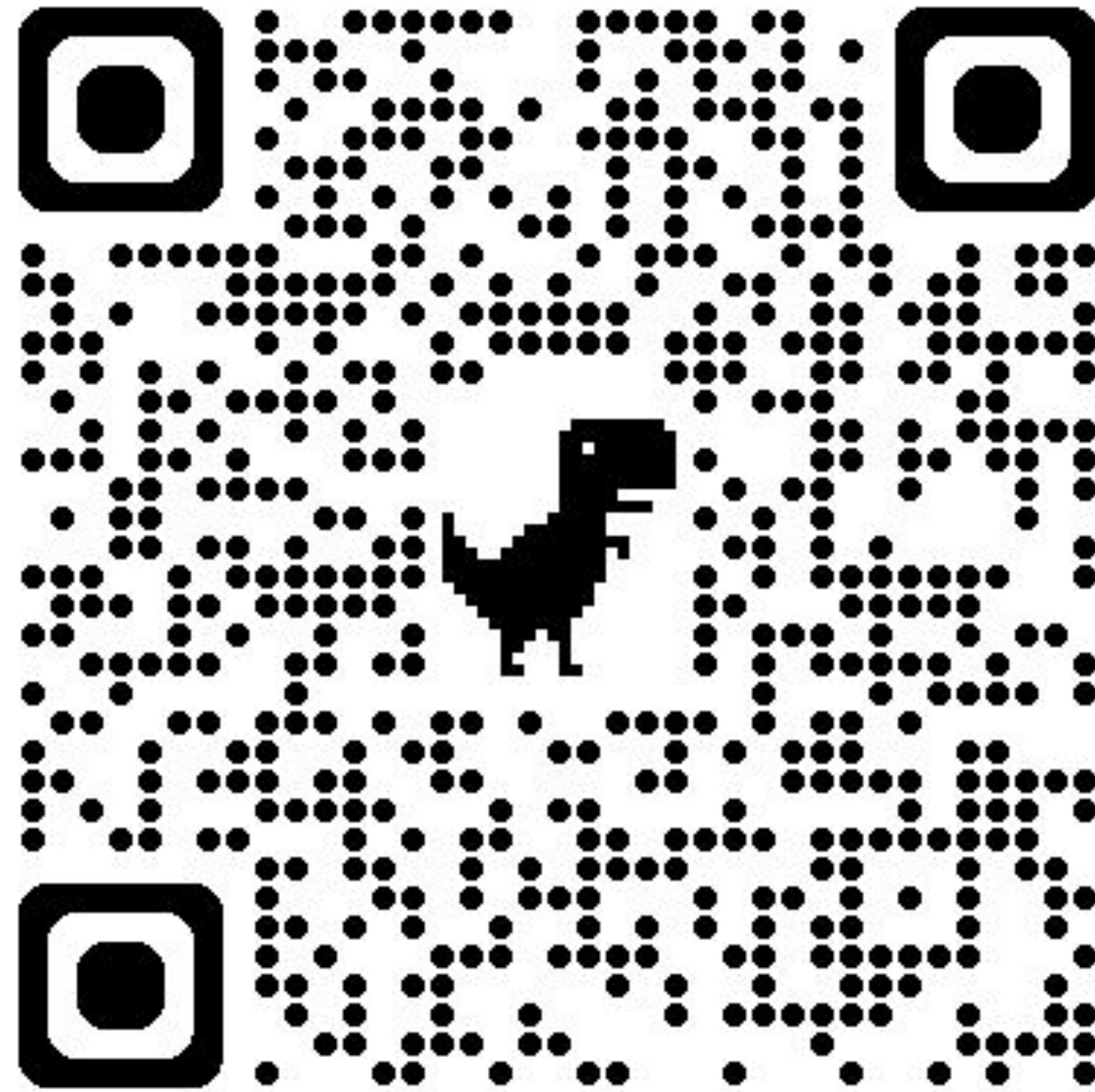


Host app renderer

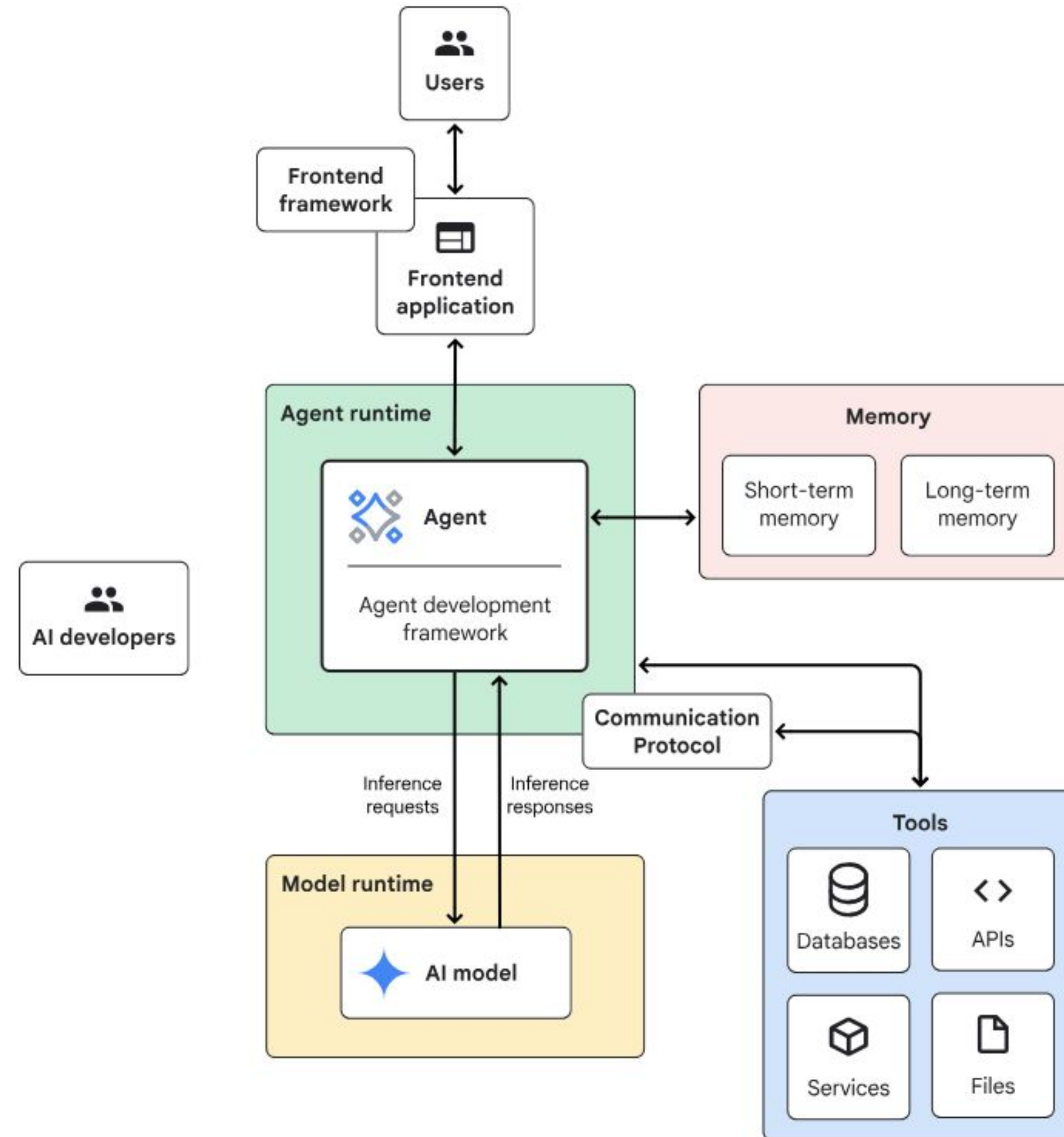
AG-UI



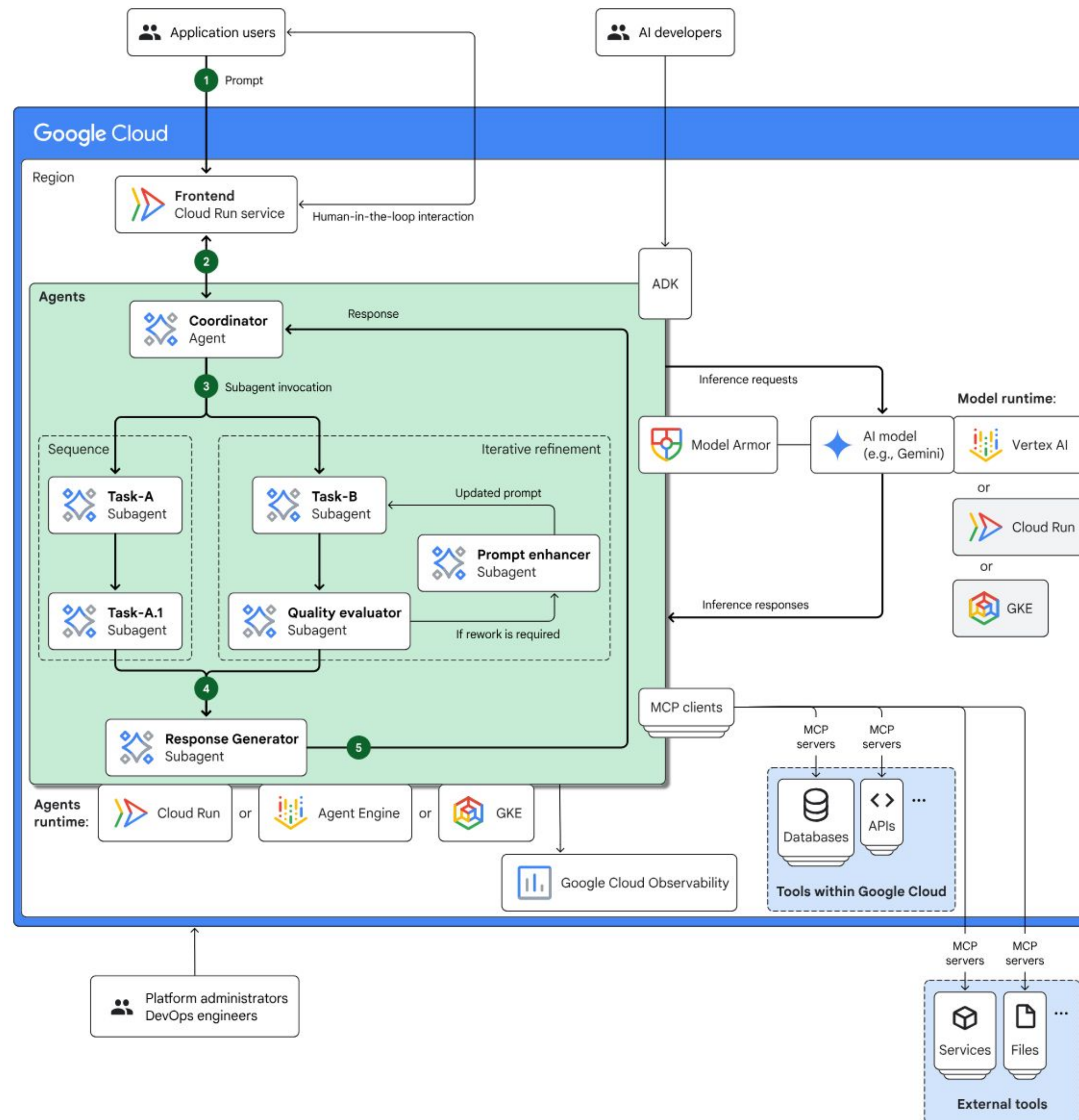
Repository



Single-Agent Architecture



Multi-Agent Architecture



Networking



LinkedIn



X



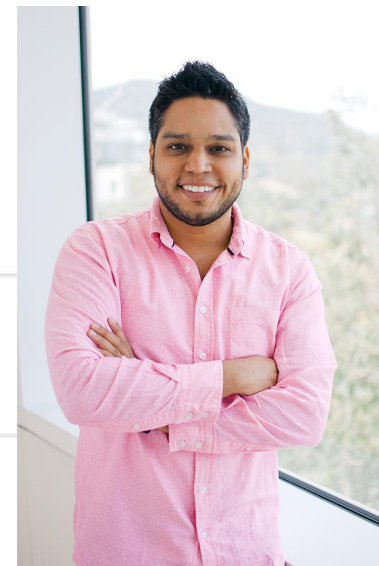
Instagram



Youtube



Thank You!
Gracias !



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@jggomezt

